

Bridging the Gap: Examining Vision Foundation Models for Optical and Radio Astronomy Applications

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ABSTRACT

Context. Vision foundation models, which have demonstrated significant potential in many multimedia applications, are often underutilized in the natural sciences. This is primarily due to mismatches between the nature of domain-specific scientific data and the typical training data used for foundation models, leading to distribution shifts. Scientific data often differ substantially in structure and characteristics; researchers frequently face the challenge of optimizing model performance with limited labeled data of only a few hundred or thousand images.

Aims. This work evaluates the performance of vision foundation models in astrophysics, with a focus on identifying the best practices for adapting these models to domain-specific datasets. We aim to establish a framework for selecting, fine-tuning, and optimizing these models for common tasks in optical and radio astronomy.

Methods. We compared multiple foundation models, including self-supervised, weakly supervised, and distillation-based architectures, across two representative optical and radio datasets and two downstream tasks. Experiments involved different fine-tuning strategies, projector heads, and data preprocessing techniques, with performance evaluated on classification and detection metrics.

Results. Using features extracted by specific foundation models improves the classification accuracy of optical galaxy images compared to conventional supervised training. Similarly, these models achieve equivalent or better performance in object detection tasks with radio images. However, their performance in classifying radio galaxy images is generally poor and often inferior to traditional supervised training results.

Conclusions. These findings suggest that selecting suitable vision foundation models for astrophysics applications requires careful consideration of the model characteristics and alignment with the specific requirements of the downstream tasks. This study demonstrates that vision foundation models can be effectively adapted to astrophysical applications, provided practitioners iterate on model selection, training strategies, and data handling. The proposed framework bridges the gap between these advanced models and the unique demands of astronomy, enabling broader adoption of deep learning in the field.

Key words. Techniques: image processing, Methods: data analysis, Radio continuum: general

1. Introduction

Large machine learning models, trained on vast amounts of data encompassing multiple domains, have in recent years enabled significant advancements in both language and image processing. Known as foundation models, these serve as a cornerstone for numerous everyday applications. Foundation models are designed either for representation learning, where the goal is to capture essential characteristics of the data, or for generative purposes, where the model attempts to create new data samples similar to the training distribution. In this work, we focus on the potential of the learned representations from vision foundation models applied to astrophysical images.

While machine learning is frequently used to manage large amounts of astrophysics image data, vision foundation models are not yet commonly employed. With the exception of ResNets pre-trained on ImageNet, most works in optical and radio astronomy employ models trained from scratch (Slijepcevic et al. 2024; Lochner & Bassett 2021;

Becker et al. 2021, e.g.). This includes networks trained for tasks such as image classification and object detection that should be able to benefit directly from the many foundation models available. A more detailed literature review will follow in Sections 3 and 4, but two major themes emerge: firstly, models trained on natural images are not believed to have learned features relevant to astrophysics images, and so are disregarded, and secondly, the desire to employ lightweight models as to not over-fit small labeled datasets precludes the use of modern foundation models, whose smallest architectures tend to have 86 million trainable parameters.

Indeed, the proliferation of foundation models in the recent years has led to what feels like an overabundance of choice to the machine learning practitioner. Each model has its strengths and weaknesses; the diversity and complexity inherent in foundation models can pose substantial hurdles for scientists who wish to apply them for domain-specific inquiries. Therefore, the primary goal of this paper is to